CS471 – Lesson 19 Solutions Key

Probability: Bayes' Rule and Base Rate Reasoning

Part 1: Guided In-Class Worksheet (Instructor-Led)

1. Probability as Uncertainty and Joint Probabilities

a. **Question:** A Space Force satellite subsystem component has a probability of 0.80 of remaining operational through the next mission segment. In one sentence, interpret this probability in an operational context.

> Solution
>
> Across many comparable mission segments, the component would be expected to remain operational about 80% of the time; the outcome of any single segment remains uncertain.

b. **Question:** For a battlefield reconnaissance system, adequate visibility occurs with probability $P(V) = 0.75$, and reliable communication given adequate visibility occurs with probability $P(C \mid V) = 0.80$. Compute the joint probability $P(V, C)$.

> Solution
>
> Apply the Product Rule:
>
> $$P(V, C) = P(V) \cdot P(C \mid V)$$
>
> $$P(V, C) = (0.75)(0.80) = \mathbf{0.60}$$

2. Bayes' Rule: Updating Beliefs Under Sensor Evidence

Scenario: An autonomous satellite diagnostics system evaluates a critical propulsion valve failure (F).

- Prior probability of failure: $P(F) = 0.10$
- Detector likelihood given failure: $P(A \mid F) = 0.80$
- Marginal probability of an alert: $P(A) = 0.20$

a. **Question:** Write the Bayes' Rule formula used to compute the posterior probability $P(F \mid A)$, identifying the role of the prior, likelihood, and marginal evidence.

Solution

$$P(F \mid A) = \frac{P(A \mid F) \cdot P(F)}{P(A)}$$

- $P(F) = 0.10$: Prior baseline belief of failure before observing the sensor.
- $P(A \mid F) = 0.80$: Likelihood of the detector firing given a true failure.
- $P(A) = 0.20$: Marginal probability of an alert sounding across all states.

b. **Question:** Calculate the numerical posterior probability $P(F \mid A)$ when the detector triggers an alert.

Solution

$$P(F \mid A) = \frac{(0.80)(0.10)}{0.20} = \frac{0.08}{0.20} = \mathbf{0.40} \quad (40\%)$$

3. Base Rate Reasoning and Independence

a. **Question:** An automated early-warning detector identifies 95% of actual incoming missile threats ($P(\text{Alert} \mid \text{Threat}) = 0.95$), but threats occur in only 1% of all operational radar sweeps ($P(\text{Threat}) = 0.01$). Explain why an alert might still yield a posterior probability far below 95%.

Solution

Threats are extremely rare (1% base rate), so false alerts generated from the much larger non-threat population (99%) outnumber true detections. The 95% value is the detector's sensitivity/likelihood ($P(\text{Alert} \mid \text{Threat})$), not the updated posterior confidence ($P(\text{Threat} \mid \text{Alert})$).

b. **Question:** Suppose diagnostic testing reveals $P(\text{Radar Fail}) = 0.30$ and $P(\text{Radar Fail} \mid \text{SATCOM Fail}) = 0.30$. Are these failure events independent or dependent? Justify your answer.

Solution

Independent, because $P(\text{Radar Fail} \mid \text{SATCOM Fail}) = P(\text{Radar Fail}) = 0.30$. Knowing the state of the SATCOM link provides zero mathematical information about the radar failure probability.

Part 2: Student Independent Practice Solutions

A. Anti-Submarine Warfare (ASW) Sonobuoy Classification

Scenario: $P(S) = 0.04 \implies P(\neg S) = 0.96$, $P(A \mid S) = 0.90$, $P(A \mid \neg S) = 0.05$.

1. **Question:** Compute Joint Probabilities:

   Solution

   a. $P(S, A) = P(A \mid S) \cdot P(S) = 0.90 \times 0.04 = \mathbf{0.0360} \quad (3.60\%)$

   b. $P(\neg S, A) = P(A \mid \neg S) \cdot P(\neg S) = 0.05 \times 0.96 = \mathbf{0.0480} \quad (4.80\%)$

2. **Question:** Marginal Evidence Probability $P(A)$:

   Solution

   Apply the Law of Total Probability:

   $$P(A) = P(S, A) + P(\neg S, A) = 0.0360 + 0.0480 = \mathbf{0.0840} \quad (8.40\%)$$

3. **Question:** Posterior Probability $P(S \mid A)$:

   Solution

   $$P(S \mid A) = \frac{P(S, A)}{P(A)} = \frac{0.0360}{0.0840} = \frac{36}{84} = \frac{3}{7} \approx \mathbf{0.4286} \quad (42.86\%)$$

   Even with a 90% detection sensitivity, an alert indicates a submarine is present only 42.86% of the time due to the 4% base rate.

B. Cyber Intrusion Detection System (IDS)

Scenario: $P(\text{APT}) = 0.002 \implies P(\neg\text{APT}) = 0.998$, $P(\text{Alarm} \mid \text{APT}) = 0.98$, $P(\text{Alarm} \mid \neg\text{APT}) = 0.01$.

1. **Question:** Compute the total marginal probability of an IDS alarm, $P(\text{Alarm})$.

   Solution

   $$P(\text{Alarm}) = P(\text{Alarm} \mid \text{APT})P(\text{APT}) + P(\text{Alarm} \mid \neg\text{APT})P(\neg\text{APT})$$

   $$P(\text{Alarm}) = (0.98)(0.002) + (0.01)(0.998)$$

   $$P(\text{Alarm}) = 0.00196 + 0.00998 = \mathbf{0.01194} \quad (\approx 1.19\%)$$

2. **Question:** Compute the posterior probability $P(\text{APT} \mid \text{Alarm})$.

Solution

$$P(\text{APT} \mid \text{Alarm}) = \frac{P(\text{Alarm} \mid \text{APT})P(\text{APT})}{P(\text{Alarm})} = \frac{0.00196}{0.01194} \approx \mathbf{0.1642} \quad (16.42\%)$$

3. **Question:** Tactical decision conclusion:

Solution

The Cyber Officer should **not** automatically sever mission-critical connections based solely on a single alert, because approximately 83.58% of triggered alarms are false positives. The IDS alert should instead trigger secondary forensic logging or automated sandbox isolation.